CURRICULUM VITÆ

GOURAV KHULLAR

CONTACT INFORMATION

B315, Physics and Astronomy Building
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First-Author+Student Publications: 16, Total: 109, h-index=33, 5572 Citations
Grants at UWashington+UPitt.: $400,000*, PI-ing 100hr of JWST spectroscopic programs*

EMPLOYMENT

Baum Postdoctoral Fellow for Innovative Astronomy *2024-*
Department of Astronomy and DiRAC Institute, University of Washington

Member, DiRAC Institute Leadership Team *2026-*

Data Science Postdoctoral Fellow *2024-*
eScience Institute, University of Washington

Visiting Scientist *2026*
International Space Science Institute (ISSI), Bern

Samuel P. Langley PITT PACC Astrophysics Postdoctoral Fellow *2022-2024*
PITT PACC and Dept. of Physics and Astronomy, University of Pittsburgh

Visiting Researcher *2021-2022*
MIT Kavli Institute (Mentor: Michael McDonald)

EDUCATION

Ph.D., Astronomy & Astrophysics • UNIVERSITY OF CHICAGO, US *May 2022*
Thesis: *Star Formation Histories in Galaxy Clusters and High-Redshift Lensed Galaxies*
Advisor: Michael D. Gladders
URA Scholar, Radix Fellow, Jerry Rao Fellow

MASt. in Astrophysics • UNIVERSITY OF CAMBRIDGE, UK *2014-15*
Master of Advanced Study, Institute of Astronomy
Thesis: *Characterizing AGN and host galaxies in the Dark Energy Survey*
Advisor: Richard McMahon
INLAKS Fellow at Fitzwilliam College

Erasmus Mundus Exchange Fellow • AALTO UNIVERSITY, FINLAND *2014*
Dept. of Physics and Metsähovi Radio Observatory

B.Tech., Engineering Physics • INDIAN INSTITUTE OF TECHNOLOGY DELHI, INDIA *2010-14*
Thesis: *Stellar Speckle Interferometry and Adaptive Optics*
Advisor: Kedar Khare

HONORS & AWARDS

- AAS Lancelot Berkeley Prize, JWST UNCOVER Collaboration (Team Member) *2027*
- ALMA Ambassador, National Radio Astronomy Observatory *2024-25*
- Baum Postdoctoral Fellowship for Innovative Astronomy, University of Washington *2024-*
- Samuel P. Langley PITT PACC Astrophysics Postdoctoral Fellowship *2022-24*
- University Research Association (URA) Visiting Scholar, Fermilab *2020, 2021-22*
- Physical Sciences Teaching Prize, University of Chicago *2020-21*
- Radix Fellow, Physical Sciences Division, University of Chicago *2020*
- Graduate Student Leadership Award, University of Chicago *2018, 2021*
- Finalist, Graduate Teaching Assistant Award, University of Chicago *2016*
- Jerry Rao Fellowship, University of Chicago *2015-2016*
- Brinson Fellowship, University of Chicago *2015*
- INLAKS Foundation Scholarship, University of Cambridge *2014-15*
- Erasmus Mundus Scholarship, Aalto University, Finland *2014*

- Outstanding Contribution Award, Co-Curricular and Academic Activities, IIT Delhi *2013*
- Semester Excellence Awards, IIT Delhi *2011, 2012, 2013*

Grants & Resources

Telescope Grants (JWST) **$250,000** UW; **$140,000** UPittsburgh

- **GO 2566** (Cycle 1), **PI** **$290,000** total
  Characterizing Stellar Mass Assembly and Physical Properties in the Brightest Galaxy in the Redshift> 5 Universe
- **GO 4125** (Cycle 2), **Co-PI** **$791,000** total; **$290,000** UW+UPittsburgh
  Galaxies Under Construction: Resolved Scaling Relations and Stellar Mass Assembly as Revealed by Lensed Star-Forming Clumps at Cosmic Noon
- **GO 11827** (Cycle 5), Co-I (PI: Mohit Bhardwaj) **$94,000** total; **$7,000** UW
  Probing the Faintest Fast Radio Burst Host Galaxy with JWST
- **GO 10329** (Cycle 5), Co-I (PI: Matthew Bayliss) **$167,000** total; **$18,000** UW
  Spatially Resolving A HMXB Stellar Population at Cosmic Noon With Strong Lensing
- **GO 5594** (Cycle 3), Co-I (PI: Guillaume Mahler) **$200,000** total; **$51,000** UW
  Survey Program: SLICE, Strong Lensing and Cluster Evolution
- **GO 6675** (Cycle 3), Co-I (PI: Hakon Dahle) **$28,000** UW
  Time delay cosmography with strong cluster lenses
- **GO 3843** (Cycle 2), Co-I **$321,000** total
  Resolving Star Formation At the Star Cluster Scale Down to 30 pc at z=2.5

Other Funding

- Team Lead – Lensing and JWST Meeting – International Space Science Institute (ISSI), Bern: Two weeks workshop with financial support (2027)
- International Space Science Institute, Bern – Visiting Scientist (4 weeks, 2026)
- DEI and Community Funding for IDEA and UChicago Astro programs **$50,000**
- University Research Association (URA) Visiting Scholars Program, Fermilab **$15,000**
- NRAO ALMA Ambassador Grant, 2024-25 **$10,000**
- Astrobites/AAS Education Grants **$7,000**

Computing Allocations

- OSPool Network: **project-specific open line for credits** *2026-*
- NSF DISCOVER ACCESS: **2.3 million credits** *2024-26*
- Center for Research Computing, Pitt: **2.5 million CPU hours** *2023-25*
- Center for Research Computing, Pitt: **1.7 million CPU hours** *2023-24*

Telescope Proposals

JWST (a subset of accepted proposals shown here)

- **GO 2566** (Cycle 1), **PI** *20.0 hours*
  Characterizing Stellar Mass Assembly and Physical Properties in the Brightest Galaxy in the Redshift> 5 Universe
- **GO 4125** (Cycle 2), **Co-PI** (with Florian) *67.8 hours*
  Galaxies Under Construction: Resolved Scaling Relations and Stellar Mass Assembly as Revealed by Lensed Star-Forming Clumps at Cosmic Noon
- **GO 10329** (Cycle 5), Co-I (PI: Matthew Bayliss) *9 hours*
  Spatially Resolving A HMXB Stellar Population at Cosmic Noon With Strong Lensing
- **GO 11827** (Cycle 5), Co-I (PI: Mohit Bhardwaj) *10 hours*
  Probing the Faintest Fast Radio Burst Host Galaxy with JWST
- **GO 5594** (Cycle 3), Co-I (PI: Guillaume Mahler) *120 hours*
  Survey Program: SLICE, Strong Lensing and Cluster Evolution
- **GO 6675** (Cycle 3), Co-I (PI: Hakon Dahle) *12 hours*
  Time delay cosmography with strong cluster lenses
- **GO 3843** (Cycle 2), Co-I (PI: Matthew Bayliss) *14.9 hours*
  Resolving Star Formation At the Star Cluster Scale Down to ∼30 pc at z=2.5

- **GO 2555** (Cycle 1), Co-I (PI: Rivera-Thorsen) *24.2 hours*
  How Do Ionizing Photons Escape the Sunburst Arc?

HST

- **GO 17912** (Cycle 32 Bridge), Co-I (PI: Bayliss) *24 orbits*
  Mapping Spatially Resolved Neutral H In Strongly Lensed Quasar Host Galaxies With Narrowband Lyman-alpha
- **GO 17728** (Cycle 32), Co-I (PI: Bayliss) *44 orbits*
  Spatially Resolving Highly Ionized Channels of Lyman Continuum and Lyman Alpha Escape on 10's of Parsecs Scales In A Strongly Lensed Galaxy
- **GO 17726** (Cycle 32, HST+JWST), Co-I (PI: Dahle) *14 HST orbits + 11.5 JWST hours*
  Time delay cosmography with strong cluster lenses
- **SNAP 17110** (Cycle 30), Co-I (PI: Setton) *409 orbits*
  Post-starbursts from DESI: Timing quenching and morphological transformation at $1 < z < 1.3$
- **Cycle 28**, Co-I (PI: Dahle) *3 orbits*
  A bright arc behind an extreme cluster lens at $z = 1.5$
- **SNAP 15307** (Cycle 25), Co-I (PI: Gladders) *115 orbits*
  Building the SPT-HST Legacy: Imaging Massive Clusters to $z = 1.5$

ALMA

- **Cycle 10**, Co-I (PI: Whitaker)
  Detecting Dust in a Sample of Lensed Quiescent Galaxies at Cosmic Noon
- **Cycle 10**, Co-I (PI: Setton)
  Does Molecular Gas Survive Quenching Near Cosmic Noon?
- **Cycle 8**, Co-I (PI: Di Mascolo)
  Unveiling the thermodynamical properties of the most distant and massive ICM selected clusters

Ground-Based

- **Gemini-S Fast Turnaround**, **PI** *1 night, 2025*
  GMOS Spectroscopic follow-up of the first lensed arc detection in Rubin/LSST Data Preview 1 imaging
- **Gemini-N and Gemini-S Fast Turnaround**, **PI** *3 nights, 2021*
  GMOS Spectroscopic follow-up of COOL-LAMPS early type galaxies
- **Apache Point Observatory ARC 3.5m**, **PI** *15 nights, 2025,2026*
  Spectroscopy of high-redshift lensed galaxies
- **Magellan/PISCO**, **PI** *2 nights, 2018*
  griz observations of SPT galaxy clusters
- **Magellan (LDSS3, IMACS, FOURSTAR and FIRE)**, Co-I *6 nights, 2020, 2021*
  Observations of high-redshift gravitationally lensed galaxies
- **Magellan/LDSS3**, Co-I *2 nights, 2017*
  Spectroscopic observations of high-redshift DES clusters

Telescope Observing

68 nights of observing experience

- Magellan Telescopes: LDSS3, IMACS, FIRE, FOURSTAR, MIKE (37 nights) *2016-24*
- APO 3.5m Telescope: KOSMOS, NICFPS (+ ARCTIC, Echelle, TripleSpec) (15 nights) *2024-*
- CTIO/Blanco Telescope: DECam (7 nights, Dark Energy Survey) *2017*
- Nordic Optical Telescope: ALFOSC (5 nights) *2019-21*
- McDonald Observatory: VIRUS-P (3 nights) *2021*
- Himalayan Chandra Telescope, Indian Institute of Astrophysics (1 night) *2012*

Professional Service & Leadership

Leadership

- Co-Founder and Survey Manager, JWST LEGGOS Collaboration *2023-26*
- Leadership Team, DiRAC Institute, University of Washington *2026-*
- Team Leader, Lensing+JWST Team, International Space Science Institute (ISSI) *2026-27*
- Lead Member, choIR Extragalactic Astronomy Team *2025-*
- Co-founder, COOL-LAMPS Collaboration *2019-*
  (ChicagO Optically-selected strong Lenses, Located At the Margins of Public Surveys)

Department and University Committees

- Equity and Inclusion Committee (Postdoc Representative), UW Astronomy *2025-26*
- Equity and Inclusion Council, A&A, UChicago *2016-19*
- Dean's Student Advisory Committee, UChicago *2015-17*
- Teaching Committee and Student Representative, IoA, University of Cambridge *2014-15*
- Academic Affairs Officer, Fitzwilliam College, University of Cambridge *2014-15*
- President, Astronomy Club, IIT Delhi *2012-13*
- Physics Representative, Academic Committee, IIT Delhi *2012-13*

Conferences, Collaboration Meetings, Workshops and Seminars Organized

- ISSI Bern Workshop on JWST and Gravitational Lensing – 2 weeks *2027*
- EXtravaGALza, UWashington's Extragalactic Astronomy Symposium *2026*
- chOIR Conference: New Data That Challenge Underlying Assumptions about Galaxy Evolution *2025*
- Pitt JWST Workshop Part II *2024*
- UNCOVER JWST Collaboration Meeting *2023*
- Pittsburgh Astro Seminar Co-organizer *2023*
- COOL-LAMPS Collaboration Meeting *2021*
- Shared Leadership and Consensus Building towards Equity and Inclusion *2021*
- IDEA (Inclusion, Diversity, and Equity in Astronomy) Week *2020*
- IDEA (Inclusion, Diversity, and Equity in Astronomy) Day *2019*
- Starting Equity and Social Justice Conversations in your STEM Community, National Science Policy Network *2019*
- UChicago Strong Gravitational Lensing Discussion Group *2018-19*

Journal Review
The Astrophysical Journal (2023-2026), *MNRAS* (2024), *Machine Learning: Science and Technology* (2025)

Proposal & Panel Review
NASA Astrophysics panels (2023-2026), NSF panel (2023), JWST Time Allocation (2024), ALMA DAPR (2024), DOE Science Graduate Student Research Program (2026), NeurIPS Machine Learning and the Physical Sciences (2022)

Other Service
Media Intern, 232nd AAS Meeting, American Astronomical Society (2018)

Presentations

Select Invited Talks

- Colloquium, Penn State Astronomy *Dec 2026*
- Galaxies Seminar, Geneva Observatory *Oct 2026*
- Institute Seminar, Institut d'Astrophysique de Paris, Sorbonne *May 2026*
- Astro Lunch Talk, UC Berkeley *Nov 2025*
- Bash Symposium, University of Texas Austin *Sep 2025*
- University of Washington Astronomy Colloquium *Apr 2025*
- Seminar, Indian Institute of Science Education and Research (IISER) Pune *Jan 2025*
- University of Chicago Astronomy & Astrophysics Seminar *Apr 2024*
- STScI Galaxies and AGN Journal Club *Nov 2023*

- McGill Trottier Space Institute Astro Seminar *Nov 2023*
- UV Galaxies Conference, Iceland *Jun 2023*
- University of Washington Astro Seminar *Apr 2023*
- Carnegie Mellon University (CMU): Impossible Problems Seminar *Apr 2023*
- Tufts University Astro Seminar *Apr 2023*
- STAtistical Methods for the Physical Sciences (STAMPS, CMU) Seminar *Mar 2023*
- Survey Science Group Seminar, University of Pittsburgh *Oct 2022*
- NOIRLab Scientific Lunch *Nov 2021*
- Harvard Galaxy Clusters Meeting *Nov 2021*
- UT Austin Extragalactic Astronomy/Cosmology Seminar Series *Nov 2021*
- Yale Galaxy Lunch *Nov 2021*
- MIT Kavli Brown Bag Lunch *Oct 2021*
- High Redshift Galaxy Evolution Meeting, Harvard CfA *Oct 2021*
- Institute for Astronomy, University of Hawaii *Sep 2021*
- University of Massachusetts Amherst Galaxy Lunch *Apr 2021*
- University of Michigan Astronomy Galaxy Meeting *Oct 2020*

Select Conference Talks

- IAUS411: Star Clusters as a Guide to Galaxies, Argentina *Nov 2026*
- Plenary Talk, JWST Through Cluster Lenses, Santander, Spain *Aug 2026*
- The End of Star Formation, UIUC *Feb 2026*
- Big Galaxies, Big Problems, Lorentz Workshop, Leiden *May 2025*
- Physics of early galaxy and black hole formation with JWST, IAP, Paris *Dec 2024*
- IAU General Assembly, South Africa, JWST Special Session IAUS391 *Aug 2024*
- The James Webb Space Telescope turns one: the birth and growth of galaxies *Jul 2023*
- European Astronomical Society Meeting, Gravitationally Lensed Galaxies *Jul 2022*
- Bayesian Deep Learning for Cosmology and Time Domain Astrophysics *Jun 2022*
- KIAA Forum on Gas in Galaxies for Early Career Scientists *Nov 2021*
- Spatially Resolved Spectroscopy with Extremely Large Telescopes (recorded) *Sep 2021*
- COOL-LAMPS Collaboration Meeting *Jun 2021*
- Multi-Object Spectroscopy for Statistical Measures of Galaxy Evolution (lightning) *May 2021*
- 237th AAS Meeting: Characterizing the brightest known galaxy in the redshift > 5 Universe *Jan 2021*
- 237th AAS Meeting: COOL-LAMPS Collaboration, A proposed model for inclusive undergraduate teaching and research *Jan 2021*
- Galaxy Formation and Evolution in the Era of Roman Space Telescope (Talk) *Oct 2020*
- South Pole Telescope Clusters Collaboration Meeting *2017-21*
- University of Chicago Astronomy Chalk Talk *Mar 2019*
- 232nd American Astronomical Society Meeting *Jun 2018*

Poster Presentations

- MIT First Light JWST Conference *Jun 2023*
- Multi-Object Spectroscopy for Statistical Measures of Galaxy Evolution *May 2021*
- 232nd American Astronomical Society Meeting *Jun 2018*
- 228th American Astronomical Society Meeting *Jun 2016*

Research Advising & Mentoring

Primary Research Advisor

- Julissa Sarmiento (PhD student, University of Pittsburgh) *2023-*
- Dylan Berry (Post-baccalaureate researcher, University of Washington) *2026-*
- Juliana Karp (PhD student and NSF Graduate Research Fellow, UWashington) *2025-26*

Research Supervisor, Graduate and Undergraduate Projects *2022-*

- Alex Ross (Undergraduate student, University of Washington) – Recipient of the Goldwater Scholarship, Mary Gates Award, shortlisted for UW Rhodes/Marshall candidacy.

- Emma Tilden Lamb-Smith (Undergraduate student, University of Washington)
- Sarah Bodansky (Graduate student, University of Massachusetts, Amherst)
- Isaac Sierra (Graduate student, University of California Davis)
- Arsh Kumaran (Masters student, University of Pittsburgh)
- Simon Mork (Graduate student, Arizona State University)
- Aidan Cloonan (Graduate student, University of Massachusetts, Amherst)
- Yasha Kaushal (PhD, University of Pittsburgh)
- Joshua Roberson (PhD, University of Cincinnati)
- Alex Navarre (PhD, University of Cincinnati –> NIST)
- Natalie Malagon (Undergraduate student, University of Chicago)

Thesis and Mentoring Committees *2024-*

- Julissa Sarmiento (PhD student, University of Pittsburgh); Advisor: Gourav Khullar + Rachel Bezanson
- Nguyen Binh (PhD student, University of Washington); Advisor: Arianna Long
- Tiara Anderson (PhD student, University of Washington); Advisor: Arianna Long
- Lainey Gammon (PhD student, University of Washington); Advisor: Arianna Long

Graduate Mentor, COOL-LAMPS Collaboration (PI: Michael D. Gladders) *2020-*

- **Mentored 19 undergraduate researchers, including (now graduate students at)**: Viraj Manwadkar (Stanford), Katya Gozman (Michigan), Ezra Sukay (Johns Hopkins), William Cerny (Yale), Yunchong Zhang (Pittsburgh), Finian Ashmead (Pittsburgh), Jorge Sanchez (New Mexico State), Michael Martinez (Wisconsin Madison), Erik Zaborowski (Ohio State), and *Isaac Sierra, Simon Mork, Aidan Cloonan (mentioned above)*

Mentor, Central American-Caribbean Bridge Program in Astrophysics (CENCA) *2022*

- Kaylan-Marie Achong (Undergraduate, University of the West Indies, St. Augustine)

Teaching

University of Washington

- ASTR 203: Galaxies (Guest lecturer) *2026*
- ASTR 482: Writing Scientific Papers (Guest lecturer) *2026*
- ASTR 541: Interstellar Medium (Guest lecturer) *2025*
- ASTR 102: Introduction to Astronomy (Guest lecturer) *2024, 2025*
- ASTR 400: Undergraduate Research Seminar (Guest lecturer) *2024*

University of Chicago

- ASTR 29001 & 29002: Field Course in Astronomy & Astrophysics (**Co-instructor**) *2020-*
- ASTR 11901 94: Physics of Stars, Immersion Program (**Lead instructor**) *Summer 2018*
- ASTR 11901 94: Physics of Stars (Teaching consultant) *2019-2021*
- ASTR 11901 94: Physics of Stars (Guest lecturer) *2017-2021*
- Yerkes Summer Institute, Space Explorers Program (Guest lecturer) *2016-2019*
- ASTR 18800: Philosophical Problems in Cosmology (TA) *Spring 2018*
- ASTR 18200: The Origin and Evolution of the Universe (TA) *Winter 2018*
- PHSC 12720: Exoplanets (TA) *Spring 2016*
- PHSC 12710: Galaxies (TA) *Winter 2016*
- PHSC 12700: Stars (TA) *Fall 2015*

Indian Institute of Technology Delhi

- Astronomy Club (Guest lecturer) *2014-2015*

Diversity, Equity & Inclusion, and Outreach

Equity and Inclusion

- **Co-founder, DAC** (Direct Action Coalition), University of Pittsburgh *2023-2024*
  - Peer education on equity, inclusion, and justice for early-career physicists and astronomers; organized Physics and Astronomy Equity Week (PEW)

- **Co-founder, IDEA** (Inclusion, Diversity & Equity in Astronomy), UChicago *2017-2022*
  - Built community and peer education for early-career astronomers; organized IDEA Day (2019) and IDEA Week (2020)
  - Secured >$12,000 in grants; advocated for >$30,000 for a department climate survey
  - Established a mentoring program and relocation support for incoming students, and coordinated working groups on anti-racist policy-making
- Teaching Race in the Core, Race and Pedagogy Working Group, UChicago *2018*

Science Communication and Outreach

- **Astrobites**, the astro-ph reader's digest
  - Ombudsperson *2021-22*
  - Co-chair, Administration Committee *2019-20*
  - Workshop Coordinator, Astrobites in the Classroom Study *2018-19*
  - Co-chair, Diversity Committee *2017-18*
  - Writer *2016-21*
- Astronomy Conversations Presenter, Adler Planetarium *2015-2020*
- Founding Member, Astronomy on Tap Chicago *2015-2017*
- Scouts Outreach and Undergraduate Journal Club Speaker, IoA Cambridge *2015*
- Co-founder and Organizer, AstroWeek outreach festival, IIT Delhi *2012, 2013*
- Outreach Speaker (galaxy formation simulations, radio astronomy), IIT Delhi *2012, 2013*

Public Press

- UW faculty, researchers and students earn recognition from the Berkeley Prize, Marconi Society and National Science Foundation, UW News
- JWST UNCOVER Collaboration to Receive 2027 Berkeley Prize, American Astronomical Society
- "My History is a Blip", The Deep Time Issue, High Country News
- Alex Ross receives a 2026 Goldwater Scholarship, University of Washington
- Research Breaking New Boundaries, The chOIR collaboration and conference, UWashington
- UChicago undergrads discover bright lensed galaxy in the early universe, UChicago News
- James Webb Space Telescope to offer humanity an unprecedented look at the Universe, UChicago News
- University Of Chicago Astrophysics Students Discover Galaxy Dating Back To Early Universe, CBS Chicago
- Graduate students recognized for exceptional teaching of undergraduates, UChicago
- PSD climate grants foster belonging while socially distanced, Physical Sciences Division, UChicago
- Starts With a Bang Podcast with Ethan Siegel
- Expand Your Perspective Podcast, Ep 1: The Universe: Black Holes, Exoplanets, and the Evolution of Stars, UChicago
- Astronomers use giant galaxy cluster as X-ray magnifying lens, UChicago News
- Astrobites White Paper, Women in Astronomy Blog
- U-M researchers confirm massive hyper-runaway star ejected from the Milky Way Disk, Michigan News
- The Physics of Toys, Yerkes Summer Institute, KICP, University of Chicago
- Spy vs. Spy, Yerkes Summer Institute, KICP, University of Chicago

Publications

First-Author and Student Publications: 16, Total Publications (NASA ADS): 109, h-index=33, 5572 Citations

Publications with significant contribution (* indicates student supervision)

[1] **Khullar, G.**; Bezanson, R.; Suess, K.; Setton, D.; et al. *"Caught Napping by JWST UNCOVER+MegaScience: Constraining bursty star formation histories and number densities of mini-quenched galaxies at redshifts 4-7"*, 2026, ADS

[2] *Oh, R.; **Khullar, G.**; Long, A.; et al. *"JWST Spectra Conclusively Show an Excess of Neutral Gas Outflows in Quiescent Galaxies at z=2-5"*, 2026, ADS

[3] **Khullar, G.**; Florian, M.; Bayliss, M.B.; Hutchison, T.A.; Welch, B.; et al. *"LEGGOS I: The JWST LEGGOS Survey – LEnsing and Galaxy Growth: Observing Substructures – Unpacks the Nature of Clumpy Star Formation and Quenching in Gravitationally Lensed Galaxies beyond Cosmic Noon"*, 2026, ADS

[4] Welch, B.; **Khullar, G.**; Hutchison, T.A.; et al. *"LEGGOS V: Direct abundances of N, O, Ne, S, and Ar in five lensed galaxies at Cosmic Noon"*, 2026, ADS

[5] *Ross, A.; **Khullar, G.**; et al. *"LEGGOS III: Mapping Star Formation and Dust in Gravitationally Lensed Galaxies with SUMAC, a UMAP and Clustering Framework"*, 2026, ADS

[6] Cleri, N.J.; Hutchison, T.A.; Welch, B.; **Khullar, G.**; et al. *"LEGGOS IV: A Shocking Lack of Evidence for Shocks at sub-kiloparsec Scales at* $2 < z < 4$*"*, 2026, ADS

[7] Mitsuhashi, I.; Suess, K.A.; Leja, J.; Bezanson, R.; Greene, J.E.; Burnham, E.; **Khullar, G.**; et al. *"UNCOVER/MegaScience Finds Uniform and Highly Bursty Star Formation at* $3 < z < 9$*, consistent with the High-Redshift UV Luminosity Function"*, 2026, ADS

[8] *Roberson, J.; Bayliss, M.B.; Gladders, M.D.; **Khullar, G.**; et al. *"Using Gravitational Lensing to Probe for Candidate Quintessential Galaxies in the Epoch of Reionization"*, 2026, ADS

[9] Ghosh, A.; Oldag, D.; Tauraso, M.; Connolly, A.J.; Ferguson, P.; Jones, D.; **Khullar, G.**; et al. *"Hyrax: An Extensible Framework for Rapid ML Experimentation and Unsupervised Discovery in the Era of Rubin, Roman, and Euclid"*, 2026, ADS

[10] Cerny, C.; Lagattuta, D.; Sharon, K.; **Khullar, G.**; et al. *"Resolving 3 Exotic Hyperbolic-Umbilic Lensing Configurations in the 'Cosmic Mantis': An Exploration of RXJ0437.1+0043 with JWST"*, 2026, ADS

[11] **Khullar, G.** *"Catching Galaxies Napping: Star Formation Histories of Post-Starburst Galaxies at Redshifts* > 4*"* (IAU Proceedings), 2025, ADS

[12] Hutchison, T.A.; **Khullar, G.**; Rigby, J.R.; Welch, B.; Florian, M.K.; Sharon, K.; et al. *"JWST & the Waz Arc I: Spatially Resolving the Physical Conditions within a Post-Starburst Galaxy at Redshift 5 with NIRSpec IFS"*, 2025, ADS

[13] McDonald, M.; **Khullar, G.**; Lagattuta, D.; Mahler, G. *"A Kiloparsec-scale Stellar Cavity in the Center of A402-BCG May Be Caused by Dynamic Interactions with an Ultramassive Black Hole"*, 2025, ADS

[14] Florian, M.; Gladders, M.D.; **Khullar, G.**; et al. *"JWST Catches a Strongly Gravitationally Lensed AGN In Transition from Type II to Type I"*, 2026, ADS

[15] Cerny, C.; Mahler, G.; Sharon, K.; Jauzac, M.; **Khullar, G.**; et al. *"Strong LensIng and Cluster Evolution (SLICE) with JWST: Early Results, Lens Models, and High-Redshift Detections"*, 2025, ADS

[16] Setton, D.J.; **Khullar, G.**; Miller, T.B.; Bezanson, R.; Greene, J.E.; et al. *"UNCOVER NIRSpec/PRISM Spectroscopy Unveils Evidence of Early Core Formation in a Massive, Centrally Dusty Quiescent Galaxy at z=3.97"*, 2024, ADS

[17] *Navarre, A.; **Khullar, G.**; Bayliss, M.B.; Dahle, H.; Florian, M.; Gladders, M.; et al. *"Resolving Clumpy versus Extended Lyα in Strongly Lensed, High-redshift Lyα Emitters"*, 2024, ADS

[18] *Mork, S.D.; Gladders, M.D.; **Khullar, G.**; Sharon, K.; et al. *"COOL-LAMPS. VII. Quantifying Strong-lens Scaling Relations with 177 Cluster-scale Gravitational Lenses in DECaLS"*, 2024, ADS

[19] *Klein, M.; Sharon, K.; Napier, K.; Gladders, M.D.; **Khullar, G.**; Bayliss, M.; et al. *"COOL-LAMPS VI: Lens model and New Constraints on the Properties of COOL J1241+2219, a Bright z = 5 Lyman Break Galaxy and its z = 1 Cluster Lens"*, 2024, ADS

[20] **Khullar, G.**; Nord, B.; Ciprijanović, A.; et al. *"DIGS: Deep Inference of Galaxy Spectra with Neural Posterior Estimation"*, 2022, Mach. Learn.: Sci. Technology, 3, 04LT04, ADS

[21] Vanzella, E.; Claeyssens, A.; Welch, B.; Adamo, A.; Coe, D.; Diego, J.M.; Mahler, G.; **Khullar, G.**; et al. *"JWST/NIRCam Probes Young Star Clusters in the Reionization Era Sunrise Arc"*, 2023, ApJ, 945, 53, ADS

[22] *Martinez, M.N.; Napier, K.A.; Cloonan, A.P.; Sukay, E.; Gozman, K.; Merz, K.; **Khullar, G.**; et al. *"COOL-LAMPS. III. Discovery of a Wide Separation Quasar Lensed by a Merging Galaxy Cluster"*, 2023, ApJ, 946, 2, ADS

[23] **Khullar, G.**; Bayliss, M.B.; Gladders, M.D.; et al. *"Synthesizing Stellar Populations in South Pole Telescope Galaxy Clusters: I. Measuring the Ages of Quiescent Members in the SPT-GMOS and SPT Hi-z Clusters"*, 2022, ApJ, 934, 177, ADS

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